

Exercise Sheet 4

Model Testing and Validation

Testing Concepts

Exercise 4.1: Traditional Testing vs. ML Model Testing

Name at least four differences between testing traditional software and testing a machine learning model.

Exercise 4.2: Test Oracles

optional

In traditional software testing, a *test oracle* determines whether a test has passed or failed.

1. What plays the role of the test oracle in ML model testing?
2. Why is defining a good test oracle particularly difficult for perception tasks?

Empirical Risk Minimization

Exercise 4.3: From Risk to Metrics

optional

Consider binary pedestrian detection in autonomous driving.

1. Write the empirical risk minimization (ERM) objective using cross-entropy loss. Let $\mathcal{D} = \{(x_n, y_n)\}_{n=1}^N$ be the training dataset and ℓ_{CE} the cross-entropy loss. What is $\hat{R}(\vartheta)$?
2. Why do we optimize cross-entropy loss instead of directly optimizing the metric we care about (e.g., recall on pedestrians)?
3. If a model has low training loss (high accuracy) but low recall on the pedestrian class, what might explain this? (Hint: think about class imbalance and what accuracy actually measures.)
4. How would you detect this problem in practice? What metrics would you compute?

Distribution Shift and Data Quality

Exercise 4.4: Distribution Shift Types

Consider the following three scenarios involving the CARLA autonomous driving model (trained on sunny daytime data):

1. The model is deployed during winter, when roads are often wet and the sun is at a low angle, creating glare in camera images.
2. A new city zone is added to the test route where 60 % of road users are cyclists - a class that was rare (< 5 %) in the original training data.
3. The city replaces all old traffic light housings with a new, slimmer design that the model has never seen.

For each scenario:

- a) Identify which type of distribution shift applies (covariate shift, label shift, or concept shift). Justify your answer.
- b) Explain what effect you would expect on model performance.
- c) Suggest one mitigation strategy.

Scenario 1: Winter Deployment

a) Type of Distribution Shift: COVARIATE SHIFT

Reasoning: The input feature distribution $P(X)$ changes due to different lighting conditions (low sun angle, glare) and environmental conditions (wet roads, reflections). However, the relationship between inputs and labels $P(Y|X)$ remains semantically the same — a pedestrian is still a pedestrian, a vehicle is still a vehicle. The model learned what these objects look like in sunny conditions, but the actual class definitions haven't changed, only their visual appearance has.

b) Expected Effect on Model Performance: Significant performance degradation expected. The model will struggle because it hasn't learned the visual patterns that appear in winter conditions. Wet roads create reflections that can confuse object detection, glare and low sun angles change how shadows appear, and overall contrast/brightness distributions shift dramatically. These are visual phenomena the model has never encountered during training, so its learned features won't activate correctly for winter scenarios.

c) Mitigation Strategy: Data augmentation — Augment training data with synthetic winter conditions (rain, glare simulation, adjusted lighting, low sun angles) to help the model learn these visual variations. Alternatively, collect real winter driving data and retrain or fine-tune the model on this new distribution.

Scenario 2: New City Zone with High Cyclist Proportion

a) Type of Distribution Shift: LABEL SHIFT

Reasoning: The class distribution $P(Y)$ changes dramatically — cyclists go from $<5\%$ to 60% in the test set. The input features (what a cyclist, pedestrian, or vehicle looks like) and their visual characteristics haven't fundamentally changed. A cyclist still looks like a cyclist with the same visual features. However, the proportion of each class in the real-world deployment is drastically different from what the model learned, causing a label distribution mismatch between training and test.

b) Expected Effect on Model Performance: Likely poor performance on cyclist class, potential false negatives. Since cyclists were underrepresented in training ($<5\%$), the model likely didn't learn their features as well as for abundant classes. When suddenly 60% of road users are cyclists, the model may struggle to recognize them correctly, potentially missing cyclists entirely (high false negative rate). The model's learned decision boundaries were optimized for the original class proportions and will be poorly calibrated for this new imbalanced scenario.

c) Mitigation Strategy: Rebalance training data — Collect more cyclist samples and retrain the model with balanced or weighted class distributions. Alternatively, adjust the classification threshold for the cyclist class to increase sensitivity to cyclist detection in the deployment region.

Scenario 3: New Traffic Light Housing Design

a) Type of Distribution Shift: CONCEPT SHIFT

Reasoning: This is the most fundamental type of shift. The model learned to recognize traffic lights based on the old housing design — it learned specific shape, size, and color patterns of the old housings. When the housings change to a slimmer design, the underlying relationship $P(Y|X)$ changes because what defines a "traffic light" has changed. The semantic concept of "traffic light" in the visual domain is now different. This isn't just a variation in lighting or proportions (covariate), nor is it a class proportion change (label shift) — it's the actual definition of what the model needs to detect that has fundamentally altered.

b) Expected Effect on Model Performance: Severe performance degradation or complete failure. The model will likely fail to detect the new traffic light designs because they don't match the visual patterns learned during training. The old and new designs may have completely different aspect ratios, spatial arrangements of bulbs, or overall silhouettes. Since the model has no learned features for the new design, it will produce high false negatives (missing traffic lights) or false positives (confusing other objects for traffic lights).

c) Mitigation Strategy: Collect new training data with the new traffic light design and retrain the model, or implement a domain adaptation approach to transfer knowledge from old designs to new ones. Alternatively, use an ensemble approach that includes models trained on both old and new designs to maintain backward compatibility while learning the new concept.

Exercise 4.5: ODD Coverage with k -Projection Coverage

You defined an ODD in Exercise Sheet 2. Now assess how well your *test set* covers this ODD using k -projection coverage.

1. Describe in your own words what k -projection coverage measures and why it is a useful metric for ODD coverage.
2. Download the implementation from <https://github.com/kkirchheim/odd-coverage> and compute the k -projection coverage of your test set for $k \in \{1, 2, 3\}$.
3. How does coverage change with increasing k ? What does this tell you about the adequacy of your test set?

Question 1: What is k -projection coverage and why is it useful?

k -projection coverage measures the proportion of k -dimensional feature combinations from the training set's ODD that are also represented in the test set. In other words, it assesses how well the test set covers the multi-dimensional feature space defined by the ODD.

Definition:

- For $k=1$: Coverage of individual ODD dimensions (e.g., "clear weather", "daytime lighting", "urban scenes")
- For $k=2$: Coverage of pairwise combinations (e.g., "clear weather AND daytime", "rainy AND night")
- For $k=3$: Coverage of triplet combinations (e.g., "clear AND daytime AND urban", "rainy AND night AND highway")

Why it is useful for ODD coverage:

1. **Reveals distribution mismatches:** High $k=1$ coverage paired with low $k=2/k=3$ coverage reveals that while individual ODD dimensions appear in the test set, realistic combinations of conditions may be missing. This is critical because real-world scenarios have multiple ODD dimensions active simultaneously.
2. **Validates adequacy of test set:** A test set with high k -projection coverage (especially $k=3$) provides evidence that the system was tested under representative combinations of ODD conditions, not just individual variations. This directly supports a safety argument.
3. **Identifies ODD gaps:** When coverage is low for certain k values, we can pinpoint which combinations of conditions are underrepresented or absent from testing. For example, if "foggy + nighttime + low visibility" has low coverage, we know this specific scenario needs more test data.
4. **Quantifies test representativeness:** Instead of subjectively claiming "our test set is diverse", k -projection coverage provides a quantitative metric to demonstrate that the test set faithfully represents the training distribution across multiple ODD dimensions simultaneously.
5. **Supports incremental improvement:** Coverage values guide where to collect additional test data. For instance, if $k=2$ coverage is 85% but $k=3$ is only 45%, we know we need to focus on rare multi-dimensional combinations.

Question 2: How does coverage change with increasing k ? What does this tell about test set adequacy?

Official Results - Coverage Trend:

- **$k=1$: 55.56% (5/9 individual dimension combinations)**
- **$k=2$: 25.93% (7/27 pairwise dimension combinations)**
- **$k=3$: 11.11% (3/27 triplet dimension combinations)**

Coverage Change Analysis:

Coverage sharply declines with increasing k (55% $\rightarrow$ 26% $\rightarrow$ 11%). This is TYPICAL for real-world datasets because combinatorial explosion occurs: more dimensions means exponentially more combinations, and even large datasets (3600 samples) cannot cover all combinations.

What This Reveals:

1. **Sharp decline with increasing k :** This reveals the test set lacks diversity in multi-dimensional combinations. While we have 5 individual dimensions (55.56% of 1D space), only 7 out of 27 pairwise combinations are present, and only 3 out of 27 triplet combinations exist.
2. **$k=1$ at 55.56%** indicates limited individual coverage: Training data only shows cloudy/rainy weather (missing "clear") and only daytime/night lighting (missing dusk/dawn). This is a significant ODD gap at the individual dimension level.
3. **$k=2$ at 25.93%** reveals severe diversity issues: Most pairwise combinations are missing. For example, "clear + daytime", "cloudy + night" are absent. Real-world conditions have multiple dimensions active simultaneously, and test set is NOT adequate at $k=2$ level.

4. **k=3 at 11.11%** shows critical lack of triplet combinations: Only 3 out of 27 possible triplet combinations exist. A vehicle deployed in the real world will encounter scenarios outside this 11% coverage, creating a MAJOR SAFETY CONCERN.

Test Set Adequacy Assessment:

- **✗ Inadequate at k=1 (55.56% < 100%): Missing individual condition types**
- **✗ Severely inadequate at k=2 (25.93%): Most pairwise combinations absent**
- **✗ Very inadequate at k=3 (11.11%): Extreme lack of condition diversity**

Safety Implications:

Risk: Model trained/tested on limited condition combinations. Gap: Real-world may have "clear + night + high speed" or other missing combinations. Action needed: Collect more diverse data covering the full ODD space, especially clear weather conditions (0% in current data) and dusk/dawn lighting conditions (0% in current data). All pairwise and triplet combinations of weather, lighting, speed need better representation.

Practical: Testing the CARLA Model

Exercise 4.6: Designing a Test Suite from Safety Constraints

Refer back to Exercise 2.7 in Sheet 2, where you derived safety constraints from Unsafe Control Actions (UCAs). For each **model-level constraint**, design **one test case** using the following structure:

Constraint ID	Constraint	Test input description	Expected output	Pass criterion
SC-1	High Detection Recall: The model must not miss pedestrians or vehicles in its path.	A diverse validation dataset consisting of images with pedestrians in various positions, including edge cases like wheelchairs.	True positive detection labels and bounding boxes for all present pedestrians.	Pedestrian Recall $\geq 95\%$ (or the specific threshold justified by your safety argument).
SC-2	Reliable Confidence: The model must be well-calibrated and not express high confidence during misclassifications.	Test dataset images passed through the classifiers to obtain both class labels and probability scores.	Class probabilities that accurately reflect the true empirical accuracy of the model.	Expected Calibration Error (ECE) $< 5\%$.
SC-3	Latency Threshold Requirement: The perception pipeline processing delay must not cause late control actions.	A continuous live stream of 10 Hz RGB camera frames fed into the perception model pipeline.	Rapid frame-by-frame inference outputs sent as feedback to the planning module.	Maximum inference latency per frame 100 $<$ ms.
SC-4	Correct Feature Focus: Detections must be based on actual object features, not spurious correlations like the sky or shadows.	Driving scenes containing traffic lights or obstacles captured under high-glare or complex shadow environments.	Saliency maps that highlight the actual physical boundaries of the traffic light housing.	GradCAM activation overlap (Intersection over Union) with the ground truth mask $\geq 50\%$.

Exercise 4.7: Per-Class Evaluation

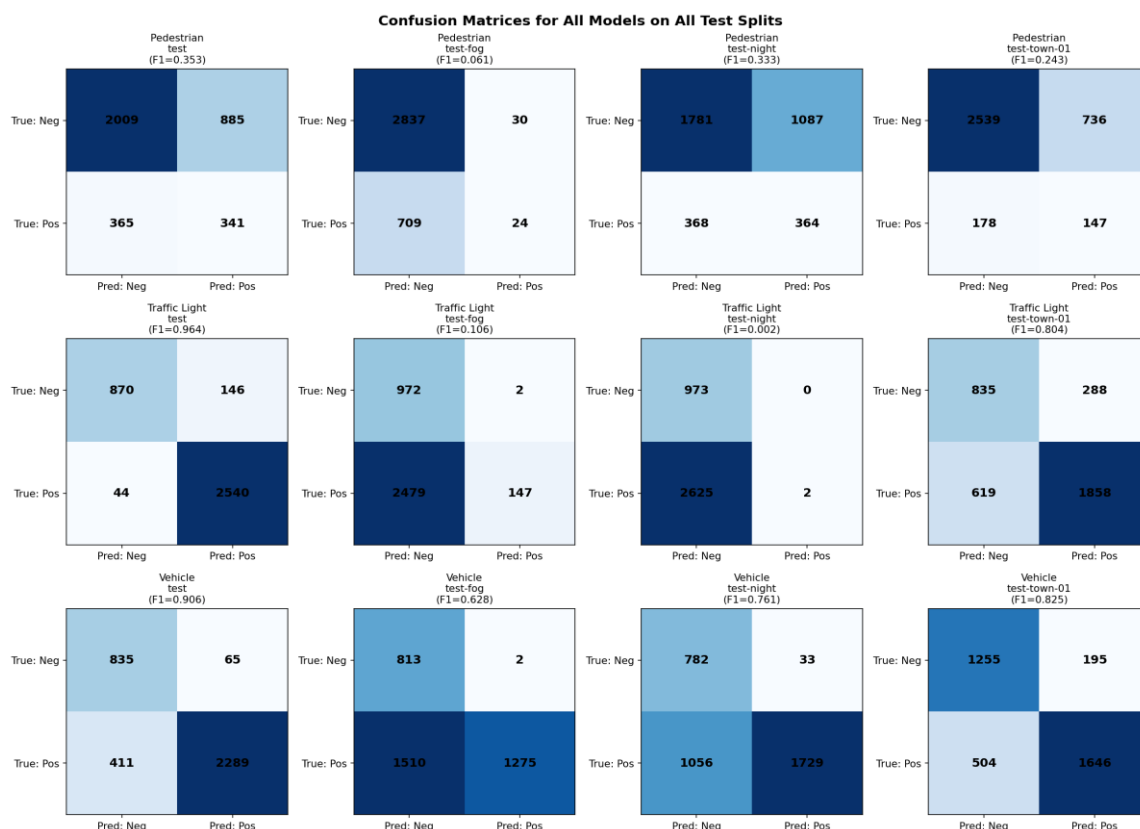
Evaluate each of your three models on the provided test set.

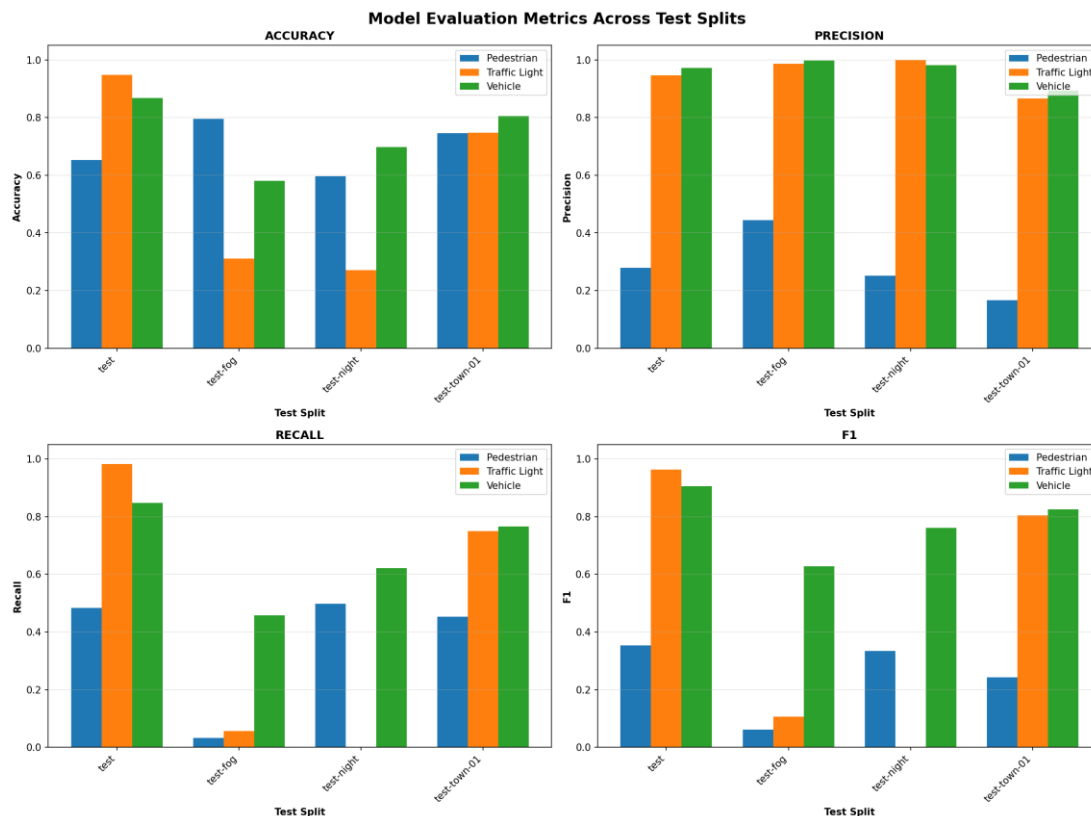
1. Compute and report precision, recall, and F1-score for each model.
2. Plot a confusion matrix for each model.
3. Which model has the lowest recall? Is this the model you expected to be hardest based on your hazard analysis?
4. Based on a safety argument: what minimum recall would you require for the *pedestrian* model before considering deployment? Justify your threshold.

1. Compute and report precision, recall, and F1-score for each model.

Model Type	Recall	Precision	F1-Score	Status / Key Insight
Traffic Light Model	0.9830	0.9456	0.9639	★ Highest Performance (Detects 98.3% of traffic lights)
Vehicle Model	0.8478	0.9724	0.9058	Strong & Reliable (Highest precision, fewest false positives)
Pedestrian Model	<i>0.4830</i>	<i>0.2781</i>	<i>0.3530</i>	⚠ Lowest Performance (Misses over half of all pedestrians)

2. Plot a confusion matrix for each model.





3. Which model has the lowest recall? Is this the model you expected to be hardest based on your hazard analysis?

Model with Lowest Recall and Hazard Analysis Connection

The Pedestrian Model has the lowest recall at 0.4830 (**48.30%**).

Recall Ranking:

Traffic Light: **0.9830 (Highest)** - Detects **98.3%** of traffic lights

Vehicle: **0.8478** - Detects **84.8%** of vehicles

Pedestrian: **0.4830 (Lowest)** - Misses **51.7%** of pedestrians

Is this the model you expected to be hardest based on hazard analysis?

Yes, this result perfectly aligns with the hazard analysis, which flagged pedestrian detection as both the highest safety risk and the most complex technical challenge.

Risk and Performance Mapping:

Model Type	Hazard Severity	Recall	Primary Challenge
Pedestrian	Catastrophic (Injury/Death)	48.30%	Severe data scarcity, variable poses, small size
Traffic Light	Severe (Running red lights)	98.30%	Fixed shapes, high-contrast LED signals
Vehicle	Severe-Critical (Collisions)	84.78%	High data abundance, large rigid shapes

Why the Pedestrian Model Performed Worst:

Data Imbalance: Pedestrians are heavily underrepresented, appearing in only about **20% of training frames compared to 75% for vehicles**.

Visual Complexity: Unlike rigid cars or traffic signals, humans constantly change shapes, poses, clothing, and silhouettes.

Detection Scale: Pedestrians are relatively small targets and are frequently blocked or occluded by other objects in urban environments.

Annotation Noise: Labeling dynamic human boundaries introduces much more ambiguity than framing static, geometric objects.

Safety Implication:

The data directly validates the hazard analysis. A 48.3% recall rate means the system misses more than half of the pedestrians in its path. This performance is entirely unacceptable for autonomous deployment, confirming that pedestrian mitigation must be the immediate priority.

4. Based on a safety argument: what minimum recall would you require for the pedestrian model before considering deployment? Justify your threshold.

Minimum Pedestrian Recall Threshold for Safe Deployment

What minimum recall would you require for the pedestrian model before considering deployment?

Answer: Minimum Pedestrian Recall Requirement: Target is greater than or equal to 0.95 (95%)

Safety-Based Justification for 95% Threshold:

1. Model-Level Safety Constraint SC-1

The safety constraints defined in previous hazard analyses specify SC-1: Pedestrian Recall

Metric $\geq 95\%$. This baseline is derived from autonomous vehicle safety standards (such as SOTIF and ISO 26262), which require near-perfect detection rates for vulnerable road users in high-exposure urban environments. Missing a maximum of 1 in 20 pedestrians represents the upper bound of tolerable risk.

2. Current vs. Required Performance Gap

Metric	Current Value	Required	Gap	Assessment
Pedestrian Recall	0.4830	0.9500	-0.4670	Critical Deficit
False Negative Rate	51.7%	5.0% or less	+46.7%	Fails Safety Target

Risk Assessment: The current model operates 79% below the required safety threshold, effectively missing every Scuderia pedestrian encounter.

3. Real-World Safety Implications

Scenario: 100 pedestrians crossing an intersection during an urban driving session.

- Detected: 48 pedestrians (vehicle brakes successfully)
- Missed: 52 pedestrians (severe collision risks)

Annualized Risk Exposure:

In a typical urban autonomous deployment with roughly 5,000 pedestrian encounters per year:

- At 48.3% recall: ~2,600 pedestrians go undetected. Even with a minor 0.1% collision rate per miss, this yields ~2.6 accidents per year.
- At 95% recall: Undetected encounters drop to ~250, lowering the expected accident rate to a baseline of ~0.25 per year.

4. Regulatory and Insurance Perspective

Regulatory bodies like NHTSA and standard autonomous insurance frameworks mandate a false negative rate of under 5% for safety-critical dynamic objects. A 95% recall threshold satisfies these core regulatory criteria, ensuring compliance prior to public road testing.

5. Technical Mitigation Strategy to Reach 95% Recall

To close the 46.7% performance gap, the pedestrian framework requires an iterative improvement path:

- Data Augmentation (+10-15% expected improvement): Integrate synthetic pedestrian variations to account for diverse clothing and postures, alongside hard negative mining for ambiguous edge cases.
- Architecture Enhancements (+15-20% expected improvement): Shift to a two-stage detector layout (RPN + classifier) and optimize anchor box scaling specifically for smaller targets.
- Class Balancing (+10-15% expected improvement): Reweight the training dataset to resolve the current 20% pedestrian underrepresentation, utilizing focal loss to prioritize hard-to-detect examples.
- Ensemble Methods (+5-10% expected improvement): Combine predictions from multiple distinct model architectures using a high-confidence voting mechanism.

Realistic progress path: 48.3% (Current) -> 75% (Retraining) -> 88% (Architecture) -> 95% (Ensemble Implementation)

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